Fiscal Year:	FY 2019	**Task Last Updated:**	FY 04/17/2019
PI Name:	Bloomfield, Susan A. Ph.D.		
Project Title:	Iron Overload and Oxidative Damage: Regulators of Bone Homeostasis in the Space Environment		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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City:	College Station	**State:**	TX
Zip Code:	77843-4375	**Congressional District:**	17
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	08/01/2013	**End Date:**	12/31/2018
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2018 per NSSC information (Ed., 6/19/18) NOTE: End date changed to 5/31/2018 per F. Hernandez/ARC (Ed., 7/31/17) NOTE: End date changed to 7/31/2017 per NSSC information (Ed., 5/5/16)		
Key Personnel Changes/Previous PI:	No changes		
COI Name (Institution):	Ford, John Ph.D. (Texas Engineering Experiment Station) Hogan, Harry Ph.D. (Texas Engineering Experiment Station) Turner, Nancy Ph.D. (Texas AgriLife Research)		
Grant/Contract No.:	NNX13AL25G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Recent work in bone biology field has demonstrated that cellular damage due to molecules called reactive oxygen species (so-called oxidative damage) is a key mechanism contributing to aging-related bone loss, as well as to that occurring with the loss of estrogen after menopause. We know very little about whether the loss of gravity as experienced in the space environment modulates oxidative damage to bone. A newly recognized source of oxidative damage in bone is that due to iron overload, or an increase in body iron stores, as has now been demonstrated in both International Space Station (ISS) crew members. Data published this month indicate that sub-pathological levels of serum ferritin (a marker for iron stores) contribute to accelerated bone loss and to incident vertebral fractures in healthy middle-aged men and women. We have preliminary data indicating that similar moderate elevations in body iron stores, as induced by a high iron diet, can result in loss of cancellous bone and diminished bone formation activity in rodents. The experiments in this proposal will utilize iron overload as an experimental intervention to exacerbate oxidative damage during a period of simulated microgravity (by hindlimb unloading in male and female mice) to define the gender-specific mechanistic role of oxidative damage as a key regulator of bone mass and, ultimately, fracture risk as modulated by the gravitational unloading of the space environment. In a final specific aim, we will test whether superimposing acute and continuous (over 42 days) radiation exposure exacerbates oxidative damage and bone decrements during hindlimb unloading. These experiments are deliberately designed to enable a rapid transition to flight experiments requiring minimal intervention during flight aboard ISS, should these ground-based simulations yield provocative findings to justify flight experiments. Iron overload can be maintained by dietary intake, and limiting the acute radiation exposure in Experiment 2 to just before the unloading period allows for testing of the same hypotheses in a flight experiment. The experiments outlined in this proposal address how microgravity affects key regulatory mechanisms governing alterations in bone (from OCB-2 in the Space Biology Plan) and also one of the National Research Council (NRC) Decadal Survey's highest priority recommendations, to expand our understanding of gender differences in adaptation to the space environment, particularly in the area of bone and long-term radiation risks.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We have evidence that oxidative damage caused by molecules that oxidize DNA or other tissue components can contribute to the loss of bone mass and integrity that occurs with aging and with the loss of estrogen in post-menopausal women. There are some unfortunate individuals who experience pathological increases in body iron stores (with diseases like hemochromatosis) who, we believe, also experience increased oxidative damage to multiple tissues due to an excess of free iron molecules. However, we now know that even moderate (sub-clinical) elevations in body iron stores in healthy middle-aged individuals can result in accelerated bone density loss. Only women in these groups, however, exhibit an increased risk (5-fold) of incident vertebral fractures. These experiments in skeletally mature mice will yield important and detailed information on the impact of these moderate elevations in body iron on evidence for oxidative damage and for accelerated bone loss, with the unique added treatment of reduced weight bearing (simulating prolonged bed rest in humans). Because this project will be examining both male and female mice, we will be able to define if there is any gender difference in the response of bone to iron overload, with and without coincident disuse. Our final specific aim will test the impact of increased iron stores in combination with disuse and exposure to chronic, very low dose gamma radiation, simulating space radiation. Radiotherapy patients on Earth experience a 2.6-fold increase in fracture risk compared to age-matched peers. A better understanding of the additional impact of radiation on bone integrity, with and without iron overload, may lead to better protection of bone health in radiotherapy patients, reducing fracture risk.
Task Progress:	Tasks accomplished (Specific Aim 2) --Year 5 (no-cost extension period) 1. Oxidative stress markers and iron status assays on stored serum and liver samples are now complete. 2. Mechanical properties analyses: Most of the Instron testing has been completed; calculations of material properties await image analysis to determine cross-sectional moment of inertia in long bone samples and computation of material property values from raw data collected during testing. This will be priority one as soon as the current academic semester ends (early May, 2019) so these data can be integrated into manuscripts. 3. Manuscript preparation: This has been the focus of our work over this last period and will continue until all results are published in the peer-reviewed literature; see below (Public Dissemination of Results) for detail. Corollary/tissue sharing projects: Given the cost and labor-intensive nature of constructing and maintaining this continuous radiation facility, we worked to maximize the data yield by actively sharing tissues with other laboratories/investigators and building on our own piggyback projects. Most extensively, my Co-Investigator, Dr. Nancy Turner, has been collecting feces and gut epithelial cells to examine changes in the gut microbiome with iron overload and/or hindlimb unloading and/or continuous irradiation (CIRR). Dr. Turner reports that further work on this awaits the start of a graduate student in her laboratory at Michigan State University. Given her laboratory's recent findings in this area, these data will confirm any independent effects of altered iron status and/or hindlimb unloading in mice, to complement our recent findings of significant changes in unloaded adult rats (see Ritchie LE et al., 2015). Abstracts/manuscripts: Results were presented at the American Society of Bone & Mineral Research 2019 annual meeting in Montreal. Two manuscripts have been published over this final period; two more will be ready for submission within 60 days, and many more are in development. Published manuscripts and presentations are listed in the below Bibliography section of this Task Book report. Manuscripts in active preparation: RS Bokhari, AM Marich, JP Elizondo, CE Metzger, E Sturgell, LJ Suva, HA Hogan, J Ford, SA Bloomfield. Bone volume changes less, but mechanical properties more, in middle-aged than in young adult female mice with hindlimb unloading and/or continuous, very low dose-rate gamma radiation. For submission to Bone. Metzger CE, Bokhari RS, Seidel D, Black J, Allen MR, Matt Allen, Hogan HA, Zwart S, Turner ND, Bloomfield SA. Sex-specific adaptations to iron overload in hindlimb unloaded mice are associated with altered serum hepcidin. For submission to FASEB J. Planned manuscripts (not all inclusive): Ford J, Saucier D, Bokhari RS, Bloomfield SA. Creating a homogeneous, very low dose-rate radiation field for continuous use in an animal facility: Monte Carlo simulations vs. dosimetry measures. for submission to Radiation

	Research. Bokhari RS, Elizondo JP, Turner ND, Ford J, Suva LJ, Hogan HA, Bloomfield SA. Does bone impact of space-relevant radiation exposure accumulated over 70 days differ from one acute exposure at the same dose? for submission to Radiation Research. Bokhari RS, Metzger CE, Seidel D, Black J, Allen MR, Matt Allen, Hogan HA, Zwart S, Turner ND, Bloomfield SA. Modeling the space environment: interactions between unloading and continuous, low-dose rate gamma irradiation. For submission to npj/Microgravity. Reference: Ritchie LE, Taddeo SS, Weeks BR, Lima F, Bloomfield SA, Azcarate-Peril M, Zwart SR, Smith SM, Turner ND. Space environmental factor impacts upon murine colonic microbiota and mucosal homeostasis. PLoS One 10 (6): e0125792, 2015.
Bibliography Type:	Description: (Last Updated: 01/20/2026)
Abstracts for Journals and Proceedings	Bokhari R, Metzger C, Marich A, Sturgell E, Allen M, Flack A, Suva L, Bloomfield S. "Adult vs.middle-aged bone responses to hindlimb unloading in males and females." Poster presentation at ASBMR 2018 (American Society for Bone and Mineral Research) Annual Meeting, Montreal, Canada, September 28-October 1, 2018. J Bone Miner Res 2018;33 (Suppl 1). , Sep-2018
Articles in Peer-reviewed Journals	Bokhari R, Metzger C, Black J, Franklin K, Boudreaux R, Allen M, Macias B, Hogan H, Braby L, Bloomfield SA. "Positive impact of low-dose, high-LET radiation on bone in partial- and/or full-weightbearing adult mice." npj/Microgravity. 2019 Jun 4;5:13. eCollection 2019. https://doi.org/10.1038/s41526-019-0074-3 ; PubMed PMID: 31231675; PubMed Central PMCID: PMC6547738 , Jun-2019
Articles in Peer-reviewed Journals	Bokhari RS, Metzger CE, Allen MR, Bloomfield SA. "Daily acute bouts of weight-bearing during hindlimb unloading mitigate disuse-induced deficits in cancellous bone." Gravit Space Res. 2018 Dec;6(2):2-11. http://gravitationalandspacebiology.org/index.php/journal/article/view/808 , Dec-2018
Dissertations and Theses	Bokhari RS. "Skeletal Impacts of Continuous Low Dose-Rate Gamma Radiation Exposure during Simulated Microgravity." Dissertation, Texas A&M University, August 2018. , Aug-2018
Dissertations and Theses	Marich AM. "Continuous, low dose irradiation results in no detriment to bone after 4, 12 and 28 days." Thesis, Texas A&M University-College Station, May 2018. , May-2018